

Cryptography

Course Information

Arkady Yerukhimovich

August 24, 2026

Course Information

- Course Title: CS 4331 / 6331 - Cryptography
- Professor: Arkady Yerukhimovich
- Class Time: Mondays 12:45-3:15pm
- Class location: 1957 E St., Room 112
- Webpage: <https://gw-cs4331.github.io/>

Outline

1 What this course is about

2 Course Details

What is Cryptography?

According to Wikipedia

Cryptography is the practice and study of techniques for secure communication in the presence of adversarial behavior.

Why Study Cryptography?

- Cryptography is amazing
 - Can do things that seem impossible

Why Study Cryptography?

- Cryptography is amazing
 - Can do things that seem impossible
- Cryptography is useful
 - Widespread applications

Why Study Cryptography?

- Cryptography is amazing
 - Can do things that seem impossible
- Cryptography is useful
 - Widespread applications
- Cryptography is beautiful
 - Deep theory
 - Definitions for core concepts
 - Everything viewed through an adversarial lens

Why Study Cryptography?

- Cryptography is amazing
 - Can do things that seem impossible
- Cryptography is useful
 - Widespread applications
- Cryptography is beautiful
 - Deep theory
 - Definitions for core concepts
 - Everything viewed through an adversarial lens

Announcement

Crypto does not mean cryptocurrency

What This Course Will and Will Not Do

Covered topics:

- Key concepts of modern crypto
- Security guarantees provided by standard crypto primitives and why we care about them
- Relations between basic crypto primitives and how to build more complex ones from simpler ones

What This Course Will and Will Not Do

Covered topics:

- Key concepts of modern crypto
- Security guarantees provided by standard crypto primitives and why we care about them
- Relations between basic crypto primitives and how to build more complex ones from simpler ones

Not covered topics:

- How to design your own crypto schemes
- How to implement crypto for real-world use
- System and network security

What This Course Will and Will Not Do

Covered topics:

- Key concepts of modern crypto
- Security guarantees provided by standard crypto primitives and why we care about them
- Relations between basic crypto primitives and how to build more complex ones from simpler ones

Not covered topics:

- How to design your own crypto schemes
- How to implement crypto for real-world use
- System and network security

Summary Course Goals

This course will not make you a cryptographer! But, it will teach you when you need to find one, and how to speak to them when you do.

What We Will Cover

	Secrecy	Integrity
Private-Key	Private-key encryption	Message authentication codes
Public-Key	Public-key encryption	Digital signatures

What We Will Cover

	Secrecy	Integrity
Private-Key	Private-key encryption	Message authentication codes
Public-Key	Public-key encryption	Digital signatures

Building Blocks:

- Pseudorandom Generators
- Pseudorandom Functions
- Hash Functions
- Number Theory

What We Will Cover

	Secrecy	Integrity
Private-Key	Private-key encryption	Message authentication codes
Public-Key	Public-key encryption	Digital signatures

Building Blocks:

- Pseudorandom Generators
- Pseudorandom Functions
- Hash Functions
- Number Theory

Key Concepts:

- Security definitions
- Adversarial mindset
- Proofs by reduction

Outline

1 What this course is about

2 Course Details

Prerequisites

The main prerequisite for this class is *mathematical maturity*

- Be able to follow mathematical definitions, theorems, proofs
- Basic logic, probability (most things will be covered in class, but some background is helpful)

Recommended prerequisite courses:

- Undergrad: CS 2312 (Discrete II), CS 3313 (Foundations)
- Grad: CS 6212 (Algorithms)

Textbook: “Introduction to Modern Cryptography, 2nd edition,” by Katz and Lindell

- This book is strongly recommended
- eBook available for free from GW Library
(https://wrlc-gwu.primo.exlibrisgroup.com/permalink/01WRLC_GWA/1piqqnm/alma99185917007604107)
- Make sure you get the 2nd edition

How to Contact Me

- The best way to contact me is
 - Via Piazza (details below and on website)
 - by email: arkady@gwu.edu
- Include CS 4331 or CS 6331 in the subject line

Course Meetings and Office Hours

- Lectures: Mondays 12:45-3:15pm, in person (1957 E St., Room 112)
- Office hours: Mondays 11:00-12:00 and Thursdays 1:30-2:30 (subject to change)
- Office hours will be held in my office (SEH 4570)

Discussions and Announcements

In Class:

- In class questions and discussion are a very important part of this class
- They also count towards your participation grade
- Ask lots of questions!

Discussions and Announcements

In Class:

- In class questions and discussion are a very important part of this class
- They also count towards your participation grade
- Ask lots of questions!

Piazza: (Link on course website)

- Piazza is for questions and general discussion about lectures, homework, and anything you are curious about but didn't ask in class.
- Student are encouraged to answer each others' questions
- Homework assignments will be announced in Piazza

Discussions and Announcements

In Class:

- In class questions and discussion are a very important part of this class
- They also count towards your participation grade
- Ask lots of questions!

Piazza: (Link on course website)

- Piazza is for questions and general discussion about lectures, homework, and anything you are curious about but didn't ask in class.
- Student are encouraged to answer each others' questions
- Homework assignments will be announced in Piazza

Gradescope: (Link on course website)

- Gradescope will be used to collect and grade homework

Important

Make sure you join the class Piazza and Gradescope

Grade Breakdown

- Exam(s) - 50% (30% for the exam with the higher score)
- Research Project - 20%
- Homework - 20%
- Participation - 10%

There will be 2 exams in this class.

- Midterm – October 19th, in class
- Final – During last week of semester, exact date TBD

Exam Policies:

- You will be allowed to bring two pages (back-and-front) of notes.
- You may not work with others on the exams.
- You may not use outside resources on the exams.

Research Project

Project Description:

- Students will carry out a research project on a crypto topic of their choice that is not covered in class.
- This is a semester-long project, and will be done in groups of up to 3 students.
- Project must focus on crypto/math, not on implementation or system security.

Research Project

Project Description:

- Students will carry out a research project on a crypto topic of their choice that is not covered in class.
- This is a semester-long project, and will be done in groups of up to 3 students.
- Project must focus on crypto/math, not on implementation or system security.

Deliverables: Each group will prepare

- 5-minute talk on the project in class
- Poster on their project and present it at an in-class poster session.
- Poster session will be held during the last week of class, exact date TBD.

Research Project

Project Description:

- Students will carry out a research project on a crypto topic of their choice that is not covered in class.
- This is a semester-long project, and will be done in groups of up to 3 students.
- Project must focus on crypto/math, not on implementation or system security.

Deliverables: Each group will prepare

- 5-minute talk on the project in class
- Poster on their project and present it at an in-class poster session.
- Poster session will be held during the last week of class, exact date TBD.

Important

Start the project early, don't ruin your Thanksgiving holiday.

Homework

Homework Details:

- Homework will come out (approximately) every week
- Homework is due (on Gradescope) by 12:45PM, on the due date (right before class).
- NO LATE HOMEWORK ACCEPTED

Homework

Homework Details:

- Homework will come out (approximately) every week
- Homework is due (on Gradescope) by 12:45PM, on the due date (right before class).
- NO LATE HOMEWORK ACCEPTED

Homework Policies:

- Homework will be graded Pass/Fail on each problem
- I will ask students each week to explain their answers in class
- You may discuss general concepts, but do your own homework
- You MUST write up your own answers. Do not copy from others.

Homework

Homework Details:

- Homework will come out (approximately) every week
- Homework is due (on Gradescope) by 12:45PM, on the due date (right before class).
- NO LATE HOMEWORK ACCEPTED

Homework Policies:

- Homework will be graded Pass/Fail on each problem
- I will ask students each week to explain their answers in class
- You may discuss general concepts, but do your own homework
- You MUST write up your own answers. Do not copy from others.

On the Use of AI Tools:

- AI tools can probably solve most of the homework – DO not do this
- Homeworks are your practice for the exams
- You can use AI to explain concepts, but make sure you understand the answer and write it up yourself

- We will have quizzes during some of the lectures
- Quizzes are not graded, but will be part of class participation
- Quizzes give me a way to quickly assess how well you understand the material

Class Participation

Class participation score will consist of the following:

- Involvement during lecture and on Piazza
- Participation in quizzes
- Questions for classmates' projects

If you are sick or cannot participate a given week, please let me know.

In Class Behavior

- Treat others with respect. We have students coming from diverse backgrounds, and I want everyone to feel welcome.
- Encourage others by asking questions and helping each other
- Do not disparage anybody

Important

Everyone will enjoy the class more if we treat each other with respect.

Next – the principles of modern cryptography